Report 4: Neural Network

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1 Network Architecture

The neural network was implemented from scratch using object oriented programming.

The implementation contains 3 classes:

- Neuron
- Layer
- NeuralNetwork

The network architecture is read from a text file.

Example:

```
3
2
2
1
```

Meaning:

- 3 layers
- 2 input neurons
- 2 hidden neurons
- 1 output neuron

Each neuron contains:

- weights
- bias
- sigmoid activation function

2 Feedforward

Each neuron computes a linear sum:

$$z = \sum x_i w_i + b$$

Then applies the sigmoid activation function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The outputs of one layer become the inputs of the next layer.
Feedforward is performed layer by layer until the final output is obtained.

3 XOR Experiment

The XOR neural network was initialized using the weights and biases from the lecture slides.

Hidden layer:

$$[-10, -10],\ b = 15$$

$$[10, 10],\ b = -5$$

Output layer:

$$[10, 10],\ b = -15$$

Inputs tested:

- (0,0)
- (0,1)
- (1,0)
- (1,1)

Results:

```
[0,0] -> 0.007
[0,1] -> 0.992
[1,0] -> 0.992
[1,1] -> 0.007
```

The neural network successfully approximates the XOR function.

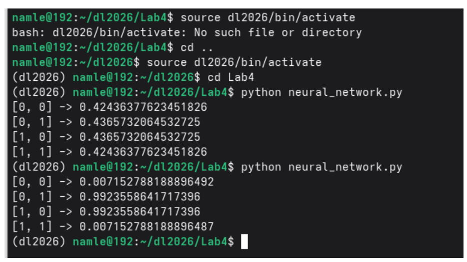

Figure 1: Results of the XOR neural network